

The Total-Span Term in a Nine-Point Certificate for Simple Critical Zeros of the Riemann Zeta Function

Li Xiang
lixiang90

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Abstract

We improve the rational certificate for a fixed perturbed Montgomery–Taylor window by using the square of its kernel at the total span of nine points. Existing point and derivative data, together with their certified convexity intervals, suffice: no new kernel table is computed. Exact difference-constraint closure and 84 closed branches replace the 47 weakest domains of an existing complete 2,399-domain cover. A standard-library rational checker verifies the entire cover, including every finite-box dual residual. The uniform local reward becomes $805403/10^8$, and the same lossless all-zero transport gives

$$\liminf_{T \rightarrow \infty} \frac{N_0^s(T)}{N(T)} \geq \frac{66812491}{99194597} = \frac{941021}{1397107} = 0.673549699486152456 \dots$$

The denominator counts all nontrivial zeros with multiplicity; the numerator counts simple zeros on the critical line. The fixed window, continuous point/region admission and analytic transport are inherited from the cited certificates. The new finite implication is checked by exact rational arithmetic. The strengthened theorem has not yet been ported to the separate Lean formalization or accepted by the record website. We make no new zero-free-region claim.

1 Statement and inherited inputs

Let $N(T)$ count all nontrivial zeros of ζ with imaginary part in $(0, T]$, including multiplicity, and let $N_0^s(T)$ count its simple critical-line zeros in that interval.

Theorem 1.1. *Within the fixed continuous certificate and proved all-zero analytic framework specified below,*

$$\liminf_{T \rightarrow \infty} \frac{N_0^s(T)}{N(T)} \geq \frac{941021}{1397107}.$$

The only new computational implication is the complete rational certificate of Section 3; no additional correlation conjecture, zero-free strip or arithmetic moment assumption is used.

The previous bound was $66812491/99194740$ [2]. The gain is

$$\frac{941021}{1397107} - \frac{66812491}{99194740} = \frac{134566003}{138585665617180},$$

approximately 0.0000970995105451 percentage points. This comparison is with the project’s fixed preceding theorem, rather than a claim of exhaustive global priority or official record acceptance.

For reproducibility we specify the unchanged window. Put $\beta = 1/\sqrt{2}$, $Z = \sqrt{2} \sin \beta$, and $I = [-1/2, 1/2]$. With zero extension outside I , define

$$f(u) = \frac{\cos(\sqrt{2}u) + \sum_{j=1}^{12} c_j \cos(2\pi ju)}{Z}, \quad K(t) = \int_I f(u) e^{-2\pi i t u} du, \quad w(t) = K(t)^2,$$

where the integer numerators $10^9 c_j$ are

$$(12310798, -15041681, 3867664, 6489926, -992327, 5580097, \\ -6846472, 3781297, -5670353, 3355089, -450523, -218483).$$

The certified density is positive, even, and has integral one. In particular $0 \leq w(t) \leq 1$ for real t . The eight-point functional is

$$F_8(h) = \sum_{r=0}^6 b_r h_r + \sum_{0 \leq i < j \leq 7} a_{ij} w(h_i + \dots + h_{j-1}),$$

where $10^8 b_r = (28898, 57272, 75526, 80958, 75526, 57272, 28898)$, so $B := \sum b_r = 404350/10^8$. The successive rows of $10^8 a_{ij}$ are

$$(13630830, 0, 19565904, 20317441, 45030671, 100000000, 200000000), \\ (29997996, 30621878, 47766489, 79682558, 109938656, 100000000), \\ (37356140, 69378121, 65335210, 79682558, 45030671), \\ (38030065, 69378121, 47766489, 20317441), \\ (37356140, 30621878, 19565904), \quad (29997996, 0), \quad (13630830).$$

There are 26 nonzero pair terms. Each fixed index-span weight sum is at most two.

Certified input 1.2. *We retain the fixed public PC8 source [1], the complete capture and continuous semantics of [2], and the majorant, energy and all-zero transport of [3, 4]. In particular:*

1. $F_8 \geq c_0 := 805003/10^8$ on $[0, \infty)^7$;
2. every separated frame with $F_8 < c_L := 805803/10^8$ lies in the union of 241 recorded closed cells and their complete reflections;
3. the original point packs certify lower values and two-sided derivative intervals of w , and the original REG endpoints certify convexity on their indicated real intervals;
4. the fixed continuous energy bound is $2 - C(f) > 67216841/10^8$.

These are disclosed inherited certificates, rather than new numerical sampling assumptions. The present finite checker does not by itself reprove their analytic admission. The preceding proportion's separate Lean proof and this strengthened rational proof have different verification states, detailed in Section 5.

2 The unused total span and valid tangents

For real gaps $g \in [4/5, \infty)^8$ set

$$F_9(g) = \frac{1}{2} \{F_8(g_0, \dots, g_6) + F_8(g_1, \dots, g_7)\} + 2w(S), \quad S = \sum_{r=0}^7 g_r. \quad (1)$$

The preceding finite certificate imposed only $w(S) \geq 0$ on its new span-eight variable. The original 2,196-point catalog already includes usable points near the weak configurations' total spans. Convexity guards, rather than new evaluations, allow us to consume this term.

Lemma 2.1 (Interval-valid reanchoring). *Suppose $v \leq w(p)$, $d_- \leq w'(p) \leq d_+$, and w is convex on an interval containing $l^* := \min(l, p)$ and $u^* := \max(u, p)$, where $l \leq u$. Then each of*

$$v + d_+(l^* - p) + d_-(x - l^*), \quad v + d_-(u^* - p) + d_+(x - u^*)$$

is at most $w(x)$ for every real $x \in [l, u]$.

Proof. Convexity gives $w(x) \geq v + d(x - p)$ with $d = w'(p)$. The tangent minus the first line is $(d - d_-)(x - l^*) + (d_+ - d)(p - l^*) \geq 0$. For the second it is $(d_+ - d)(u^* - x) + (d - d_-)(u^* - p) \geq 0$. The proof also covers points outside the smaller query interval. $\square$

Put $SC = 32768$ and $E = 10^9$. A catalog entry with integer point P and pack $V + 2^{32}D_+ + 2^{64}D_-$ supplies

$$p = P/SC, \quad v = V/10^{10}, \quad d_- = (D_+ - 2E)/E, \quad d_+ = (2E - D_-)/E.$$

Every new direct-region line is checked against that entry's original PTL list index and its original convex region. With $k = (P + 16384) \gg 15$ and region endpoints A_k, B_k , the direct guard is

$$k < 16, \quad A_k/SC \leq \min(l, p) \leq p \leq \max(u, p) \leq B_k/SC, \\ D_+, D_- \geq 1, \quad D_+ + D_- \leq 4E.$$

An alternative old-enabled line must match the point and pack of an actually enabled old atom and lie inside its already valid interval. Its inherited **TV**al assertion supplies the exact tangent lower bound throughout that interval. The same derivative-interval comparison in Lemma 2.1 proves both reanchored lines without asserting convexity beyond an original region. A value/derivative pack without either full interval guard is rejected. Closure endpoints are kept as exact rationals; they are not truncated to grid points. All lines at one physical distance constrain the same square variable.

3 Complete closed branches and rational duals

The unchanged 482 reflected labels have 482^2 ordered combinations. Shared gap bounds leave 4,796; shared spans leave 2,405; full exact difference-constraint closure leaves exactly 2,399. Negative cycles certify empty intersections. Prefix coordinates are real: using integer coefficients in units $5SC$ does not restrict configurations to a lattice. The inherited capture's direction exclusions, all 41 outer/root checks and reflected half-spaces are retained.

Replace exactly the 47 domains whose inherited lower bound is below $805403/10^8$. In each, partition its total-span interval at the interior original region endpoints. Consecutive branches are closed and share their common endpoint. Their union equals the entire original interval. Recompute full difference closure after adding each branch's two span constraints, and use the resulting projections both for the gap box and for tangent guards. There are 84 branches in total.

For example, labels 424/215 have $S \in [12.905426025390625, 13.262969970703125]$. Splitting at the original region endpoint $434048/SC = 13.24609375$ yields exact branch lower bounds approximately 0.008064135118335435 and 0.009937463795419915. The small upper branch is already paid by the old terms and pressure, without extending convexity beyond its certified interval. Labels 131/389 increase from approximately 0.0080526095277502 to 0.00811870169978014. These decimals illustrate localization only; the proof consumes the stored fractions.

Each branch has eight gap variables and 34 shared square variables, including $z_{0,8} = w(S)$. Write its constraints $Ax \leq b$ and its complete finite box $L \leq x \leq U$, with square bounds $[0, 1]$. The integer objective $q^t x$ is exactly $2 \cdot 10^8 F_9$.

Lemma 3.1 (Full residual payment). *For any rational $\lambda \geq 0$, put $r = q + A^t \lambda$. Every feasible real x satisfies*

$$F_9 \geq \frac{-\lambda^t b + \sum_i r_i (L_i \text{ if } r_i \geq 0; U_i \text{ otherwise})}{2 \cdot 10^8}.$$

Proof. The identity $q^t x = r^t x - \lambda^t A x$ gives $q^t x \geq r^t x - \lambda^t b$. Minimize each residual coefficient over its full box. No exact stationarity is required. $\square$

The actual all-zero Hermitian operator A retains all off-line conjugate-reflection pairs and multiplicities. Its remainder after the n simple critical-line columns has positive inertia p with $N - n \geq 2p$. The threshold trace argument [3, 5] gives

$$\operatorname{tr} A^2 \geq 2(N - n) + \operatorname{tr} \varphi_2(U) = 2N - n + J(U).$$

The unconditional smoothed correlation formula is $\operatorname{tr} A^2 = (C(f_\epsilon) + o(1))N$ [6, 7], and the full normalized height span is at most $N + o(N)$. Taking first $T \rightarrow \infty$ for fixed ϵ , then $\epsilon \rightarrow 0$, in (3) gives

$$\liminf \frac{n}{N} \geq \frac{2 - C(f) - B}{1 - c}.$$

The certified energy of this exact window is

$$C(f) = \frac{1}{2} + \beta \cot \beta + \frac{\sum_j c_j^2 (1/2 - 1/(4\pi^2 j^2))}{2 \sin^2 \beta}, \quad 2 - C(f) > 67216841/10^8.$$

Hence the displayed bound is at least

$$\frac{67216841 - 404350}{10^8 - 805403} = \frac{66812491}{99194597} = \frac{941021}{1397107},$$

proving Theorem 1.1. The inherited dyadic-to-cumulative argument preserves these count definitions. This proof does not replace off-line zeros by real Gram points or discard their inertia.

5 Reproduction and verification status

The public data are the following two files, together with the unchanged expanded certificate:

```
output/am-ninth-span-certificate.json
output/am-ninth-span-point-catalog.json
```

From a clean checkout run

```
python scripts/am_ninth_span_certificate.py --check
```

The checker uses the Python standard library, checks dependency identities, reconstructs all 2,399 domains and 84 closed branches, and includes all residuals. The small catalog contains mathematical point/region data, not new formalization code. It identifies the original public source by fixed commit and SHA-256. With that exact source available, an optional `-source` argument additionally reextracts every consumed entry and compares its original list index, pack and region. The offline run checks a finite rational implication under the explicitly inherited continuous point/region admission; a data hash alone does not prove an analytic kernel assertion.

The source's exact identity, completed checks and an independent mathematical review are archived with this paper. No floating solver status is accepted as a certificate, and optimized Python must refuse execution because assertions are essential to the check.

The formalization repository contains the proof of the previous bound,

$$66812491/99194740.$$

Its verification is not automatically inherited by the new branches and reward. A full Lean port of this improvement, website resource compliance and official acceptance remain uncompleted. This manuscript records the strengthened mathematical certificate and its exact scope as of October 10, 2026.

References

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- [2] Li Xiang, *Additional Admitted Tangents for Joint Nine-Point Certificates and Simple Critical Zeros of the Riemann Zeta Function*, October 9, 2026, fixed manuscript.
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